

RETAIL SALES ANALYTICS

Using Excel,Sql,Python and Power-BI

PROJECT DOCUMENTATION

Analyzing and Visualizing Retail Sales Performance

Domain: Retail Sales & Inventory Analytics

Prepared by	Murugesan V
Institution	Vinsup Skill Academy, Coimbatore
Domain	Retail Sales & Inventory Analytics
Dataset	Retail Sales Data
Tools Used	Excel,Sql,Python,Power-Bi Desktop

ABSTRACT

Retail businesses generate a large volume of transactional data every day. Extracting useful business insights from this data is essential for improving operational efficiency, customer satisfaction, and overall profitability.

The Retail Sales Analytics project focuses on developing a complete end-to-end analytics solution using Microsoft Excel, SQL, Python, and Power BI. The project begins with cleaning and preparing raw retail sales data in Excel. The cleaned dataset is then imported into a relational SQL database, where business queries are executed to analyze sales performance, customer purchasing behavior, inventory levels, and staff performance.

Python is used to perform Exploratory Data Analysis (EDA) and customer segmentation using RFM (Recency, Frequency, Monetary) analysis. Finally, Power BI is used to create interactive dashboards that present meaningful business insights through charts, KPIs, maps, and slicers.

The project enables management to monitor business performance, identify high-value customers, evaluate product demand, analyze store performance, and make data-driven decisions.

PROJECT OVERVIEW

Introduction

Retail organizations manage thousands of customer transactions every day. These transactions contain valuable information regarding customer preferences, product demand, store performance, staff productivity, and inventory status. However, without proper analysis, this information cannot be effectively utilized for business growth.

The objective of this project is to build a complete Retail Sales Analytics solution that transforms raw transactional data into meaningful business insights. The project integrates multiple analytical tools, including Microsoft Excel, SQL, Python, and Power BI, to create an end-to-end analytics pipeline.

Microsoft Excel is used for data cleaning and preprocessing. SQL is used to store and manage relational data while performing business analysis through structured queries. Python is used for Exploratory Data Analysis (EDA) and customer segmentation using RFM analysis. Finally, Power BI is used to design interactive dashboards that support business decision-making.

BUSINESS NEED

In today's competitive retail industry, organizations require accurate and timely information to improve operational performance and maximize profitability.

Without proper analytics, businesses often face several challenges, including:

- Difficulty in tracking sales performance across multiple stores.
- Limited understanding of customer purchasing behavior.
- Poor inventory management resulting in stock shortages or excess inventory.
- Inability to identify top-performing products.
- Lack of insights into staff performance.
- Time-consuming manual reporting processes.

To overcome these challenges, businesses require an integrated analytics solution capable of processing large volumes of transactional data and presenting meaningful insights through interactive dashboards.

The Retail Sales Analytics project addresses these business needs by combining Excel, SQL, Python, and Power BI into a unified analytics workflow.

PROBLEM STATEMENT

Retail organizations generate large amounts of sales and customer data from daily business operations. Although this data contains valuable business information, it often remains underutilized due to inconsistent formatting, missing values, duplicate records, and fragmented storage systems.

Manual analysis of such large datasets is inefficient and prone to errors.

The primary objective of this project is to design an end-to-end analytics solution capable of:

- Cleaning raw sales data.
- Organizing data into a relational SQL database.
- Performing business analysis.
- Segmenting customers.
- Building interactive Power BI dashboards.
- Supporting management with actionable insights.

OBJECTIVES

The primary objectives of this project are:

- Clean and preprocess raw retail sales data.
- Eliminate duplicate and inconsistent records.
- Create a structured relational database using SQL.
- Perform business analysis using SQL queries.
- Analyze customer purchasing behavior using Python.
- Calculate customer RFM metrics.
- Segment customers based on purchasing patterns.
- Design interactive Power BI dashboards.
- Generate meaningful business insights.
- Support management in making data-driven decisions.

TOOLS AND TECHNOLOGIES USED

Tool	Purpose
Microsoft Excel	Data Cleaning & Preparation
MySQL	Database Management
SQL	Data Querying & Analysis
Python	Data Analysis & RFM
Pandas	Data Manipulation
NumPy	Numerical Operations
Matplotlib	Visualization
Power BI	Dashboard Development

DATASET DESCRIPTION

The Retail Sales Analytics dataset consists of multiple relational tables that represent various business operations within a retail organization. These tables are interconnected using primary and foreign keys, ensuring data consistency and integrity throughout the analytics process. The

dataset provides detailed information about customers, orders, products, inventory, stores, staff members, brands, and product categories, enabling comprehensive business analysis.

The primary tables used in the project are:

- Customers
- Orders
- Order Items
- Products
- Categories
- Brands
- Stocks
- Stores
- Staffs

These tables collectively support sales analysis, inventory tracking, customer behavior analysis, staff performance evaluation, and dashboard reporting.

EXCEL DATA CLEANING AND PREPARATION

Introduction

Microsoft Excel was used as the first stage of the Retail Sales Analytics project. Before importing the dataset into SQL, it was essential to clean and organize the raw data. Raw datasets often contain duplicate records, missing values, inconsistent formats, and incorrect data types, which can affect the accuracy of analysis.

The objective of the Excel preprocessing stage was to improve data quality, ensure consistency across all tables, and prepare the dataset for seamless integration into the SQL database. A well-prepared dataset reduces errors during data import and provides a strong foundation for subsequent analysis using SQL, Python, and Power BI. These preprocessing tasks align with the project workflow defined for the Retail Sales Analytics project.

Objectives of Excel Data Preparation

The Excel preprocessing stage was carried out with the following objectives:

- Improve the overall quality of the dataset.
- Remove duplicate records.
- Handle missing or incomplete values.
- Standardize column names and formats.
- Validate numerical and categorical data.
- Create calculated fields required for business analysis.
- Prepare CSV files for SQL import.
- Ensure data consistency across all tables.

By completing these objectives, the dataset became reliable and suitable for advanced analytical processing.

Dataset Overview

The Retail Sales Analytics project contains multiple interconnected datasets representing different aspects of retail business operations.

The dataset includes the following tables:

Table Name	Description
Customers	Customer personal and location information
Orders	Customer order details
Order Items	Product-level transaction details
Products	Product information
Categories	Product category details
Brands	Brand information
Stocks	Inventory available in stores
Stores	Store information
Staffs	Staff details handling customer orders

Each worksheet was individually cleaned and validated before being exported into CSV format.

Standardizing Column Headers

One of the first preprocessing activities involved standardizing column names. Inconsistent column naming can create problems while importing data into SQL databases and establishing relationships between tables.

All column headers were converted into a consistent naming convention using lowercase letters and underscores.

Example

Original Name	Standardized Name
Customer ID	customer_id
First Name	first_name
Order Date	order_date
Product Name	product_name
Store ID	store_id

This naming convention improves readability and simplifies SQL query writing.

Removing Duplicate Records

Duplicate records increase redundancy and produce inaccurate analytical results. Duplicate customer information or duplicate transaction records may lead to incorrect revenue calculations.

Excel's **Remove Duplicates** feature was used to identify and eliminate duplicate rows based on unique identifiers such as Customer ID, Order ID, and Product ID.

Benefits

- Improved data integrity.
- Eliminated repeated records.
- Prevented incorrect sales calculations.
- Increased database accuracy.

Handling Missing Values

Missing values are common in business datasets. Blank fields reduce data quality and affect business analysis.

The dataset was carefully examined using:

- Filter
- Go To Special
- Conditional Formatting

Missing values were handled using appropriate methods.

Examples

Field	Action Taken
ZIP Code	Filled with "NA"
Phone Number	Filled with "NA"
Optional Fields	Replaced with default values
Mandatory Fields	Verified manually

Proper handling of missing values ensured the dataset remained complete and reliable.

Data Type Conversion

Correct data types are essential for accurate calculations and database operations.

Several columns were converted into appropriate formats.

Column	Data Type
Order Date	Date
Required Date	Date
Shipped Date	Date
Quantity	Number
List Price	Currency
Discount	Percentage

Formatting dates and numeric values correctly helped avoid SQL import errors and improved calculation accuracy.

Data Validation

To improve consistency, Excel Data Validation was applied to selected fields.

Dropdown lists were created for categorical columns such as Order Status.

Available Values

- Pending
- Processing
- Shipped
- Delivered
- Cancelled

Benefits of Data Validation:

- Reduced typing errors.
- Improved consistency.
- Prevented invalid entries.
- Simplified data entry.

Creating Derived Columns

Additional calculated columns were created to support business analysis.

The most important derived field was **Total Price**.

Formula

Total Price = (List Price × Quantity) – Discount

This calculated field represents the total revenue generated from each order item and was later used for SQL analysis and Power BI dashboards.

Other derived values included:

- Total Sales
- Revenue
- Discount Amount
- Net Sales

These calculated columns reduced computation time during analysis.

Lookup Functions

Lookup functions were used to retrieve related information from different worksheets.

Functions used include:

- VLOOKUP()
- XLOOKUP()

Example

Using Product ID to retrieve:

- Product Name
- Category
- Brand
- Price

Benefits:

- Reduced duplicate information.
- Improved consistency.
- Faster data retrieval.
- Easier maintenance.

Pivot Table Analysis

Excel Pivot Tables were created to obtain a quick summary of business performance before importing the data into SQL.

Sample Pivot Reports included:

- Total Sales by Category
- Sales by Store
- Monthly Sales
- Product-wise Revenue
- Customer-wise Purchases

Pivot Tables provided an initial understanding of business trends and helped verify data quality before advanced analysis.

Sorting and Filtering

Sorting and filtering techniques were used to identify unusual records and potential data quality issues.

Examples include:

- Highest priced products.
- Lowest priced products.
- Products with low inventory.
- Orders with large discounts.
- High-value customers.

These checks ensured that outliers and anomalies were identified before database import.

Preparing CSV Files

After completing all preprocessing activities, each worksheet was exported separately as a CSV file.

Generated files included:

- customers.csv
- orders.csv
- order_items.csv
- products.csv
- categories.csv
- brands.csv
- stocks.csv
- stores.csv
- staffs.csv

These CSV files served as the source files for SQL database import.

Benefits of Excel Data Cleaning

The Excel preprocessing stage offered several advantages.

- Improved data accuracy.
- Reduced duplicate records.
- Enhanced consistency.
- Easier SQL import.
- Better reporting quality.
- Faster business analysis.
- Improved dashboard reliability.

Without proper preprocessing, subsequent SQL queries and Power BI dashboards could produce inaccurate results.

Outcome of Excel Stage

The Excel data preparation phase successfully transformed raw retail sales data into a clean, structured, and reliable dataset. Duplicate records were removed, missing values were handled,

data formats were standardized, and calculated fields were created to support business analysis.

The cleaned dataset was exported as CSV files and imported into the SQL database for further processing. This preprocessing stage established a strong foundation for SQL querying, Python-based analysis, customer segmentation, and Power BI dashboard development, ensuring that the insights generated in later stages were accurate and dependable.

SQL Tasks Performed

The following SQL tasks were completed as part of the project:

Task 1 – Create Tables

Created all database tables based on the relational schema using **CREATE TABLE** statements.

Outcome: A structured relational database was successfully created.

Task 2 – Import Data

Imported cleaned CSV files into MySQL using the Import Wizard.

Outcome: All records were successfully loaded into the database.

Task 3 – Order Details Analysis

Used **INNER JOIN** to combine Orders, Order Items, and Products tables to display complete order information.

Business Benefit: Provides detailed transaction reports.

Task 4 – Total Sales by Store

Calculated total sales generated by each store using **GROUP BY** and **SUM()**.

Business Benefit: Helps compare store performance.

Task 5 – Top Selling Products

Identified the top 5 products based on quantity sold using **ORDER BY** and **LIMIT**.

Business Benefit: Helps identify high-demand products.

Task 6 – Customer Purchase Summary

Calculated total orders, quantity purchased, and revenue for each customer.

Business Benefit: Helps identify valuable customers.

Task 7 – Customer Segmentation

Customers were categorized into **Low**, **Medium**, and **High** spending groups using SQL conditions.

Business Benefit: Supports targeted marketing strategies.

Task 8 – Staff Performance

Calculated total revenue generated by each staff member.

Business Benefit: Evaluates employee performance.

Task 9 – Stock Alert

Listed products with stock quantity below 10 units.

Business Benefit: Helps prevent stock shortages.

Task 10 – Customer Segments Table

Created a **customer_segments** table to store the segmentation results generated from Python.

Business Benefit: Enables integration between Python and Power BI.

Outcome

SQL successfully transformed the cleaned data into a structured relational database. Business queries generated meaningful insights into sales performance, customer behavior, inventory levels, and staff productivity. The SQL database also served as the data source for Python analysis and Power BI dashboard development.

Python Data Analysis and Customer Segmentation

Introduction

Python was used to perform Exploratory Data Analysis (EDA) and customer segmentation after importing data from the SQL database. Libraries such as Pandas, NumPy, and Matplotlib were used to analyze sales data and identify customer purchasing patterns.

Data Analysis (EDA)

The following analysis was performed using Python:

- Loaded data from SQL into Pandas DataFrames.
- Checked dataset information using `info()`.

- Generated statistical summaries using `describe()`.
- Identified missing values.
- Analyzed customer purchases and sales trends.
- Created basic charts for data visualization.

Business Benefit:

EDA helped understand the dataset and identify important sales patterns before customer segmentation.

Customer Segmentation (RFM Analysis)

Customer segmentation was performed using the **RFM model**, which evaluates customer purchasing behavior based on three factors:

- **Recency (R):** Number of days since the customer's last purchase.
- **Frequency (F):** Total number of orders placed.
- **Monetary (M):** Total amount spent by the customer.

Customers were grouped into segments such as:

- High Value Customers
- Regular Customers
- New Customers
- Low Value Customers

Business Benefit:

Customer segmentation helps businesses identify loyal customers, improve marketing strategies, and increase customer retention.

Export to SQL

The final customer segmentation results were exported back to the SQL database as the **customer_segments** table.

This table was later connected to Power BI for visualization and report filtering.

Outcome

Python successfully analyzed customer purchasing behavior and generated RFM-based customer segments. These insights enhanced business decision-making and provided valuable input for the Power BI dashboards.

Power BI Dashboard Development

Introduction

Power BI was used to transform the analyzed data into interactive dashboards. The dashboards provide clear visual insights into sales performance, customer behavior, inventory status, and staff performance, enabling management to make informed business decisions.

Dashboard Features

The Power BI dashboard includes:

- Total Revenue KPI
- Total Orders KPI
- Active Customers KPI
- Average Order Value KPI
- Monthly Sales Trend
- Top Selling Products
- Sales by Store
- Staff Performance
- Customer Segments
- Low Stock Alert

Interactive slicers were added for:

- Order Status

- Product Category
- Store
- Customer Segment

Business Insights

The dashboard helped identify:

- Best-performing products.
- Highest revenue-generating stores.
- Customer purchasing patterns.
- Staff performance.
- Products with low stock.
- Revenue contribution by customer segment.

These insights support better business planning and operational decision-making.

Outcome

The Power BI dashboard successfully converted raw business data into meaningful visual reports. Interactive charts, KPIs, and filters made it easier to monitor sales performance and customer trends while supporting data-driven business decisions.

Business Insights

Based on the analysis performed using SQL, Python, and Power BI, the following key business insights were identified:

- Sales performance varied across different stores.
- A small group of customers contributed significantly to total revenue.
- Some products consistently ranked among the top-selling items.
- Staff performance differed based on the number of orders handled.
- Certain products had low stock levels and required timely replenishment.

- Customer segmentation helped identify loyal, regular, and new customers.

These insights can help management improve sales strategies, inventory planning, and customer engagement.

Results

The Retail Sales Analytics project was successfully completed using Excel, SQL, Python, and Power BI.

Key Achievements

- Cleaned and prepared retail sales data using Excel.
- Created a relational database in MySQL.
- Performed SQL queries to analyze sales and customer data.
- Conducted Exploratory Data Analysis (EDA) using Python.
- Generated customer segments using RFM analysis.
- Developed an interactive Power BI dashboard with KPIs and visualizations.
- Generated meaningful business insights to support decision-making.

Future Enhancements

The project can be further improved by implementing the following features:

- Real-time database integration.
- Sales forecasting using Machine Learning.
- Demand prediction for inventory management.
- Customer recommendation system.
- Automated ETL pipeline.
- Cloud database integration.
- Mobile-friendly Power BI reports.

These enhancements can improve business efficiency and provide more advanced analytical capabilities.

Conclusion

The **Retail Sales Analytics Using Excel, SQL, Python, and Power BI** project successfully demonstrated an end-to-end data analytics workflow. The project began with data cleaning in Excel, followed by database creation and analysis in SQL, customer analysis using Python, and interactive dashboard development in Power BI.

The project provided valuable insights into sales performance, customer behavior, staff productivity, and inventory management. By integrating multiple analytical tools, the project supports data-driven decision-making and helps businesses improve operational efficiency and customer satisfaction.

Overall, this project strengthened practical knowledge of data cleaning, database management, data analysis, visualization, and business intelligence, while demonstrating how analytics can solve real-world retail business challenges.

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